

Flash and Authorize PHY6222

Version: 20240614

Online Version

Contents

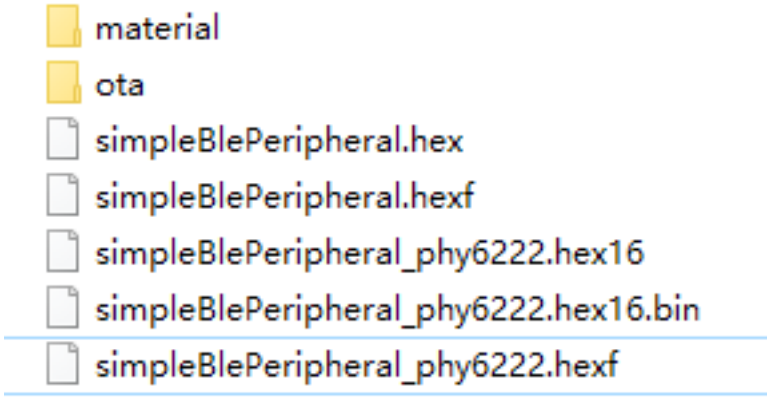
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1 Firmware flashing

Flash firmware to PHY6222 through UART using the chip vendor’ s tool.

Breakdown of PHY6222 firmware:



- `simpleBlePeripheral_phy6222.hexf`: the full firmware, used for production.
- `simpleBlePeripheral_phy6222.hex16.bin`: the firmware update, used for OTA update.

> Check the README file in the demo folder before getting started.

1.1 Tools

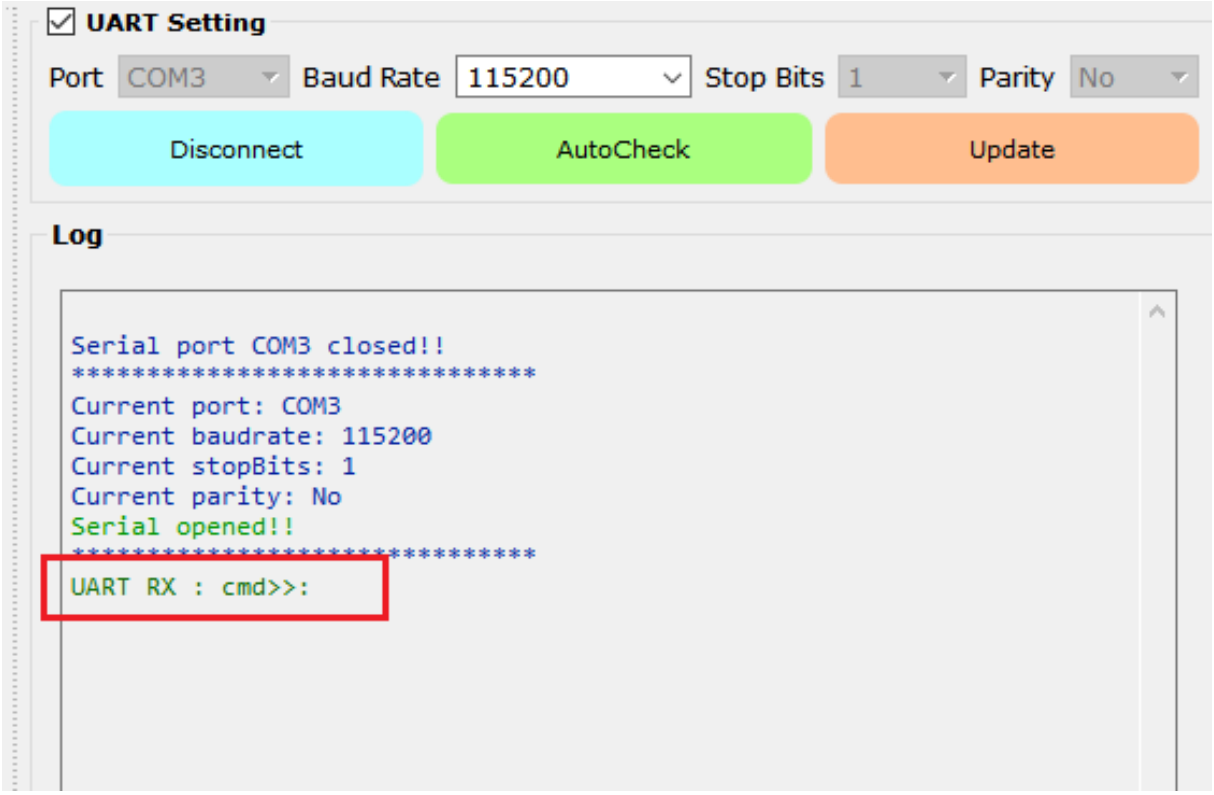
- PHY6222 development board
- USB to TTL converter
- PhyPlusKit software.

1.2 Firmware flashing

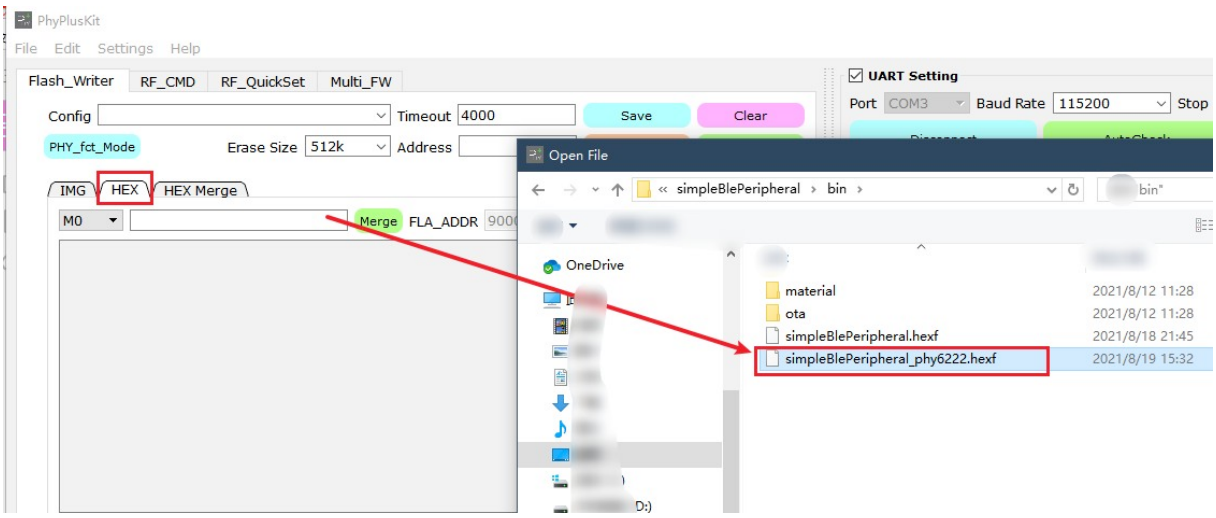
Pull pin 8 (TM) high and power on the device to make the UART receive commands. UART configuration: 115200 baud rate, 8 bit, 1 stop bit, none parity, and no flow control.

Procedure

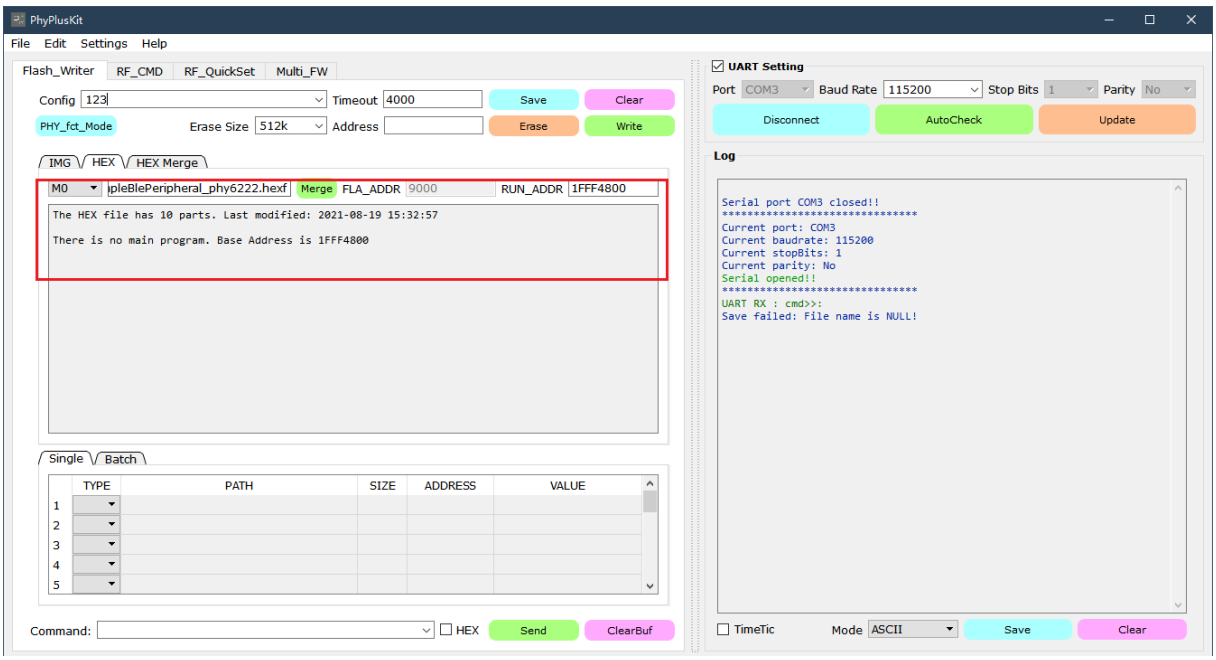
1. Prepare the software and tool and connect the hardware. The default UART for flashing is P10-RX and P09-TX.
2. Open PhyPlusKit.exe, check UART Setting, configure parameters (115200 baud rate, 8 bit, 1 stop bit, none parity, and no flow control), and click **Connect**.
3. Set TM to VDD high level, press reset on the development board, and wait for message `UART RX : cmd>>:` on the screen.



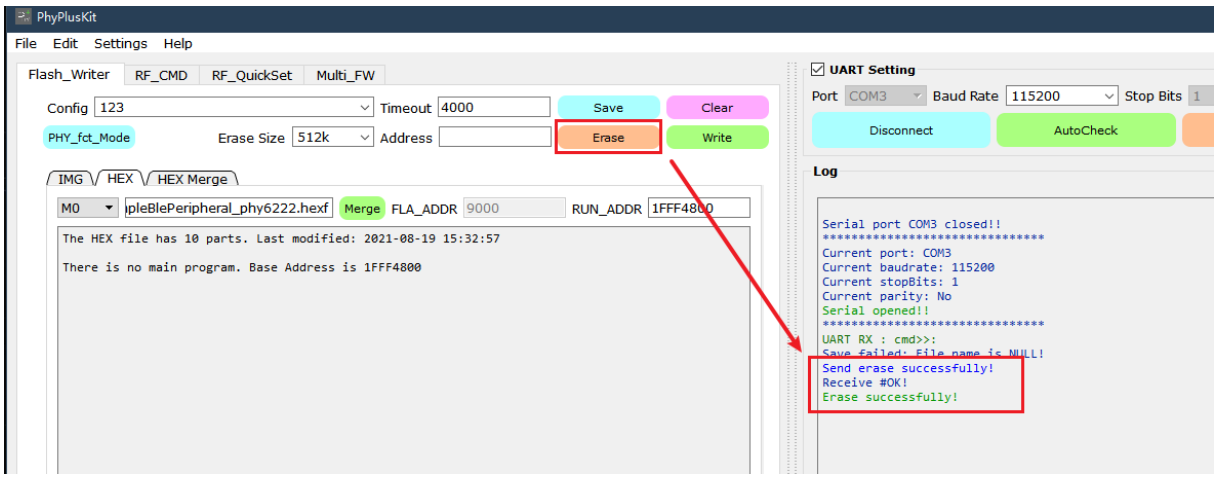
1. Double-click the input box under the HEX option to import the `.hexf` firmware file.



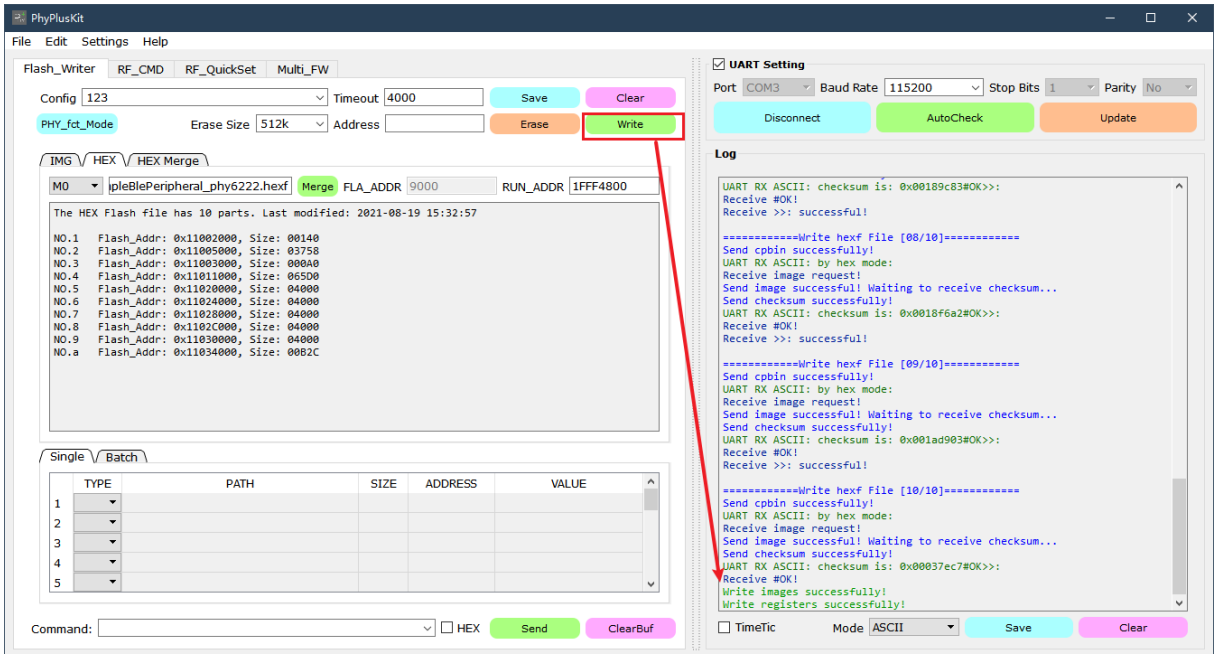
1. The program automatically parses the `.hexf` file and displays the last modification time.



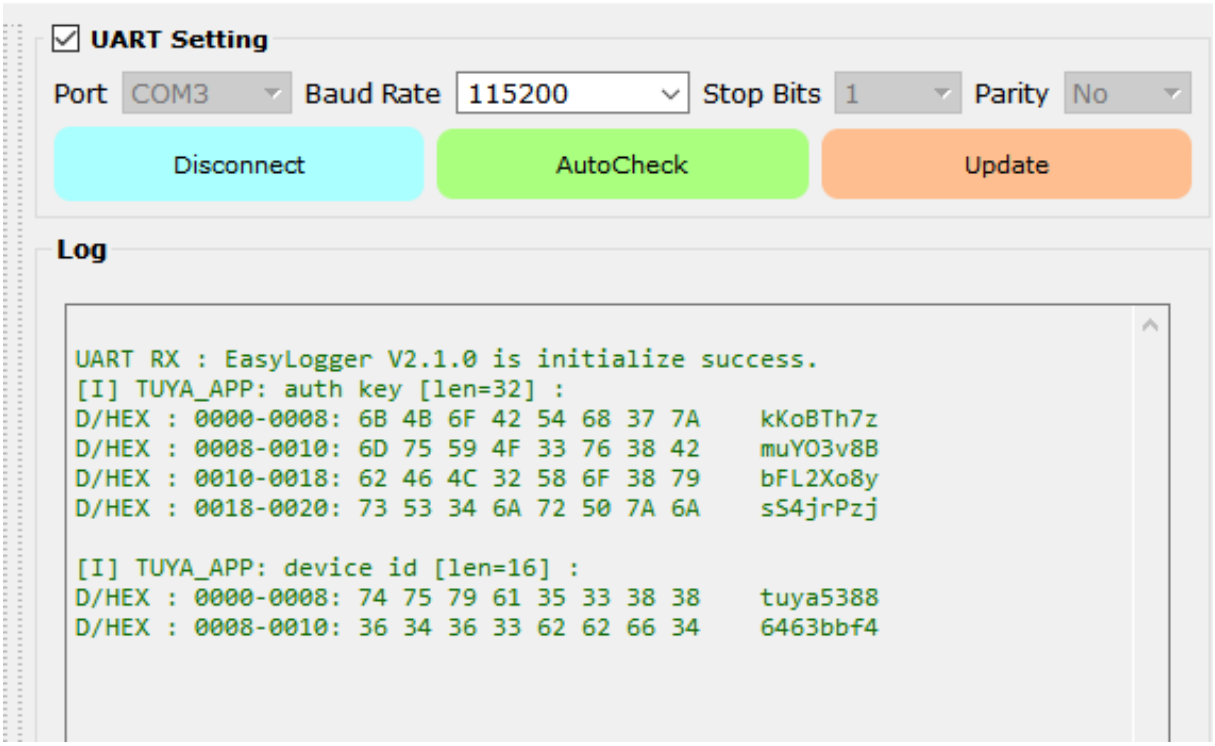
1. Before flashing, click **Erase**.



1. Click **Write** to automatically flash the `.hexf` file.



1. Set TM to GND, exit the flashing mode, press the reset button on the development board, and check the log output.



To output logs, enable the following two macros.

```
#define TUYA_BLE_LOG_ENABLE                 1
#define TUYA_APP_LOG_ENABLE                 1
```

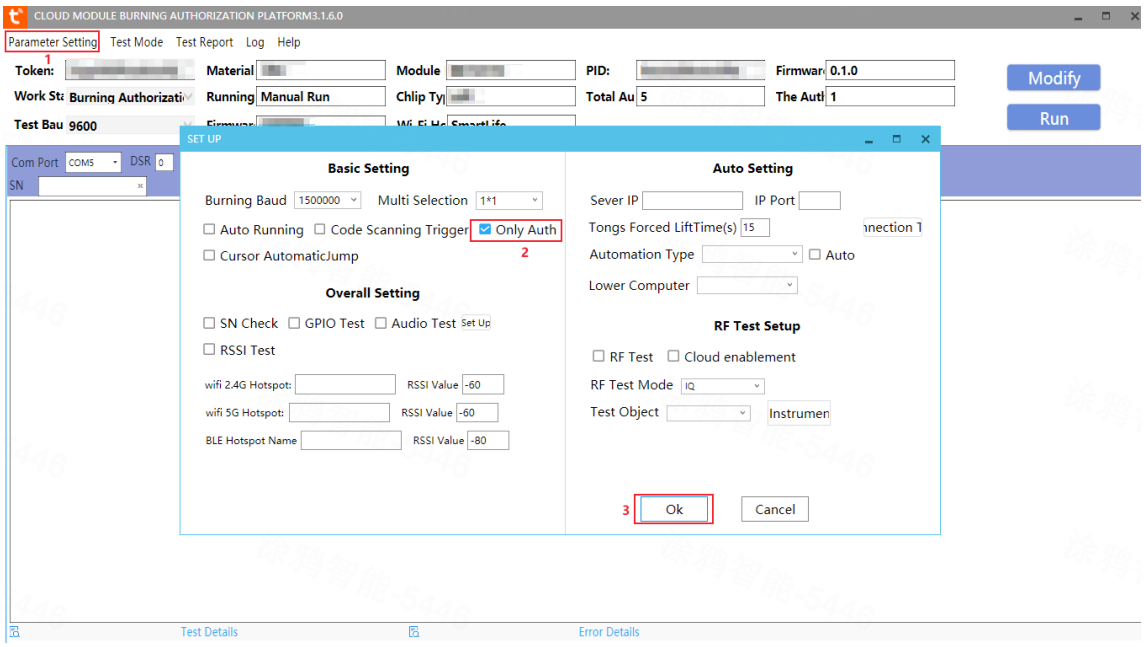
2 Firmware authorization

2.1 Get the license

For more information, see Authorize Bluetooth LE Series Modules.

2.2 Flashing and authorization

1. Connect the chip to a USB to TTL converter (RX-P26, TX-P25).
2. Open Cloud Module Burning Authorization Platform, and enable Only Auth.



> The code specifies the baud rate. Ensure that you set the software according to the code.

```c
uint32_t ty_uart2_init(void)
{
    uart_Cfg_t cfg = {
        .tx_pin = P25,
        .rx_pin = P26,
        .rts_pin = GPIO_DUMMY,
        .cts_pin = GPIO_DUMMY,
        .baudrate = 115200,
        .use_fifo = TRUE,
        .hw_fwctrl = FALSE,
        .use_tx_buf = TRUE,
        .parity     = FALSE,
        .evt_handler = phy_uart_evt_hdl,
    };
    hal_uart_deinit(UART1);
    hal_uart_init(cfg, UART1);
    hal_uart_set_tx_buf(UART1, s_uart_tx_buf, TY_UART_TX_BUF_SIZE)
        ;
    return 0;
}
```